

Statistical Inference for Linear Stochastic Approximation with Richardson-Romberg Extrapolation

Abstract

Your abstract.

1. Introduction

Stochastic approximation (SA) algorithms are a cornerstone of modern computational statistics, optimization, and reinforcement learning. Introduced by Robbins and Monro (1951), these iterative procedures provide a principled way to find roots of equations or optimize objectives when only noisy observations are available. A particularly important subclass is the *linear stochastic approximation* (LSA) algorithm, which arises naturally in temporal-difference (TD) learning, policy evaluation, and stochastic gradient descent for linear models.

In this work, we study the LSA recursion with a *constant step size* $\alpha > 0$:

$$\theta_k^{(\alpha)} = \theta_{k-1}^{(\alpha)} - \alpha \{A(Z_k)\theta_{k-1}^{(\alpha)} - b(Z_k)\}, \quad k \geq 1,$$

where $\{Z_k\}_{k \in \mathbb{N}}$ is a time-homogeneous Markov chain on a measurable space $(Z, \mathcal{Z})$ with transition kernel Q and unique invariant distribution π . The mappings $A : Z \rightarrow \mathbb{R}^{d \times d}$ and $b : Z \rightarrow \mathbb{R}^d$ are measurable functions satisfying $\bar{A} := \int_Z A(z) d\pi(z)$ and $\bar{b} := \int_Z b(z) d\pi(z)$. Assuming that $-\bar{A}$ is a *Hurwitz matrix* (all eigenvalues have strictly negative real parts), the target parameter $\theta^* = \bar{A}^{-1}\bar{b}$ is uniquely defined.

The *Polyak–Ruppert averaging* procedure (Polyak, 1990; Ruppert, 1988) provides an effective variance reduction technique. Given a burn-in period $n_0 \geq 0$, the averaged iterate is defined as

$$\bar{\theta}_n^{(\alpha)} = \frac{1}{n - n_0} \sum_{k=n_0}^{n-1} \theta_k^{(\alpha)}.$$

1.1. Bias of constant step-size iterates

The use of a constant step size $\alpha > 0$ offers several practical advantages: it enables geometrically fast forgetting of the initial condition (Dieuleveut, Durmus, and Bach, 2020) and simplifies hyperparameter tuning compared to diminishing step-size schedules. However, unlike the classical regime $\alpha_k \rightarrow 0$ with $\sum \alpha_k = \infty$ and $\sum \alpha_k^2 < \infty$, a constant step size produces iterates that converge only *in distribution* to a stationary measure Π_α , rather than almost surely to θ^* . The stationary expectation $\mathbb{E}[\theta_\infty^{(\alpha)}]$ is generally *biased* with respect to θ^* , and this bias cannot be eliminated by Polyak–Ruppert averaging alone.

As shown in Levin, Naumov, and Samsonov (2025) and Huo, Chen, and Xie (2023), the asymptotic bias admits a power-series expansion in α :

$$\lim_{n \rightarrow \infty} \mathbb{E}[\theta_n^{(\alpha)}] = \theta^* + \alpha \Delta + O(\alpha^2),$$

where $\Delta = \bar{A}^{-1} \sum_{k=1}^{\infty} \mathbb{E}[\{Q^k \tilde{A}(Z_{\infty})\} \epsilon(Z_{\infty})]$ depends on the correlation structure of the Markov chain, and $\tilde{A}(z) = A(z) - \bar{A}$ is the centered matrix-valued function.

1.2. Richardson–Romberg extrapolation

To eliminate the leading $O(\alpha)$ bias term, we employ the *Richardson–Romberg* (RR) *extrapolation* procedure. Two LSA sequences are run *on the same Markov chain trajectory* $\{Z_k\}$ with step sizes α and 2α , and the RR iterate is formed as

$$\bar{\theta}_n^{(\alpha, \text{RR})} = 2\bar{\theta}_n^{(\alpha)} - \bar{\theta}_n^{(2\alpha)}.$$

Since both sequences share the same noise realization, the leading bias term $\alpha\Delta$ cancels, leaving a residual bias of order $O(\alpha^{3/2})$ or higher (Levin et al., 2025).

More generally, one can consider the multi-level extrapolation with M step sizes $\mathcal{A} = \{\alpha_1, \dots, \alpha_M\}$ and coefficients $\{h_m\}$ determined by the Vandermonde system (Huo et al., 2023):

$$\sum_{m=1}^M h_m = 1, \quad \sum_{m=1}^M h_m \alpha_m^l = 0, \quad l = 1, \dots, M-1,$$

which reduces the bias from $O(\alpha)$ to $O(\alpha^M)$.

1.3. Problem statement and goals

The high-order moment bounds for the RR iterate $\bar{\theta}_n^{(\alpha, \text{RR})}$ established in Levin et al. (2025) show that the leading error term scales as $\sqrt{\text{Tr } \Sigma_{\epsilon}^{(M)}} \cdot n^{-1/2}$, where $\Sigma_{\epsilon}^{(M)}$ is the Markovian noise covariance defined in Section 1.4 below, matching the minimax optimal rate. Berry–Esseen type bounds and bootstrap inference procedures for the *standard* Polyak–Ruppert average $\bar{\theta}_n$ (without extrapolation) under Markovian noise have been obtained in Samsonov, Sheshukova, Moulines, and Naumov (2025).

However, the *distributional approximation* — specifically, the central limit theorem and Berry–Esseen bounds — for the *averaged Richardson–Romberg iterates* $\bar{\theta}_n^{(\alpha, \text{RR})}$ remains an open problem. The main goal of this work is to fill this gap by establishing:

1. A *central limit theorem* for $\sqrt{n}(\bar{\theta}_n^{(\alpha, \text{RR})} - \theta^*)$, identifying the limiting covariance matrix.
2. *Non-asymptotic Berry–Esseen bounds* for the rate of convergence to the Gaussian limit.
3. A practical *inference procedure* (confidence intervals) based on the RR-extrapolated iterates.

1.4. Setting and assumptions

We now formalize the setting and state the assumptions that will be used throughout this work.

Let $\{Z_k\}_{k \in \mathbb{N}}$ be a Markov chain on a complete separable metric space $(Z, \mathcal{Z})$ with transition kernel Q .

Assumption 1 (Uniform geometric ergodicity). The kernel Q admits a unique invariant distribution π and is *uniformly geometrically ergodic*: there exists $t_{\text{mix}} \in \mathbb{N}^*$ such that for all $k \in \mathbb{N}^*$,

$$\Delta(Q^k) := \sup_{z, z' \in Z} \frac{1}{2} \|Q^k(z, \cdot) - Q^k(z', \cdot)\|_{\text{TV}} \leq (1/4)^{\lfloor k/t_{\text{mix}} \rfloor}.$$

Equivalently, there exist constants $\zeta > 0$ and $\rho \in (0, 1)$ such that $\sup_z \|Q^k(z, \cdot) - \pi\|_{\text{TV}} \leq \zeta \rho^k$ for all $k \geq 1$.

Assumption 2 (Hurwitz condition and boundedness). The matrix $-\bar{A}$ is Hurwitz, i.e., all eigenvalues of $\bar{A}$ have strictly positive real parts. Moreover,

$$C_A := \max \left(\sup_{z \in Z} \|A(z)\|, \sup_{z \in Z} \|\tilde{A}(z)\| \right) < \infty,$$

where $\tilde{A}(z) := A(z) - \bar{A}$.

Assumption 3 (Noise regularity). The noise function $\epsilon(z) = \tilde{A}(z)\theta^* - \tilde{b}(z)$, where $\tilde{b}(z) = b(z) - \bar{b}$, satisfies

$$\|\epsilon\|_\infty := \sup_{z \in Z} \|\epsilon(z)\| < +\infty.$$

Under Assumptions 1–3, the error $\theta_k^{(\alpha)} - \theta^*$ satisfies the recursion

$$\theta_k^{(\alpha)} - \theta^* = (I - \alpha A(Z_k))(\theta_{k-1}^{(\alpha)} - \theta^*) - \alpha \epsilon(Z_k).$$

1.5. Key quantities

The *Markovian noise covariance matrix* captures both the marginal variance and the temporal correlations of the noise:

$$\Sigma_\epsilon^{(M)} = \mathbb{E}_\pi [\epsilon(Z_0)\epsilon(Z_0)^\top] + 2 \sum_{\ell=1}^{\infty} \mathbb{E}_\pi [\epsilon(Z_0)\epsilon(Z_\ell)^\top].$$

This matrix is the limiting covariance in the Markov chain CLT for the partial sums $n^{-1/2} \sum_{t=0}^{n-1} \epsilon(Z_t)$ (cf. Douc et al., 2018, Theorem 21.2.10).

The *asymptotically optimal covariance matrix* is given by

$$\Sigma_\infty = \bar{A}^{-1} \Sigma_\epsilon^{(M)} (\bar{A}^{-1})^\top.$$

This covariance is optimal in the Hájek–Le Cam sense and represents the best achievable asymptotic variance for estimating θ^* via averaged LSA.

The *Lyapunov equation* plays a central role in the contraction analysis. For any $P = P^\top \succ 0$, there exists a unique $Q = Q^\top \succ 0$ satisfying $\bar{A}^\top Q + Q \bar{A} = P$. Defining $a = \lambda_{\min}(P)/(2 \|Q\|)$ and $\kappa_Q = \lambda_{\max}(Q)/\lambda_{\min}(Q)$, the key contraction property holds: for all $\alpha \in [0, \alpha_\infty]$,

$$\|I - \alpha \bar{A}\|_Q^2 \leq 1 - \alpha a.$$

Since the iterates $\theta_k^{(\alpha)}$ alone are generally not Markovian (due to the Markovian noise), we consider the *joint process* $\left(\theta_k^{(\alpha)}, Z_{k+1}\right)$ with kernel

$$\bar{P}_\alpha f(\theta, z) = \int_Z Q(z, dz') f(F_{z'}(\theta), z'),$$

where $F_z(\theta) = (I - \alpha A(z))\theta + \alpha b(z)$. Under Assumptions 1–3, this joint chain admits a unique invariant distribution Π_α for sufficiently small $\alpha > 0$ (Levin et al., 2025).

2. Zeroth-Order Richardson–Romberg Difference

2.1. LSA Error Decomposition

We consider the recursion

$$\theta_k = \theta_{k-1} - \alpha_k (A(Z_k)\theta_{k-1} - b(Z_k)), \quad \alpha_k = \alpha = \text{const.}$$

Define the transition products

$$\Gamma_{m:k} = \prod_{l=m}^k (I - \alpha A(Z_l)).$$

We also introduce

$$J_k^{(0)} = - \sum_{l=1}^k \alpha \Gamma_{l+1:k} \epsilon(Z_l), \quad H_k^{(0)} = - \sum_{l=1}^k \alpha \Gamma_{l+1:k} (A(Z_l) - \bar{A}) J_{l-1}^{(0)},$$

with initial conditions $J_0^{(0)} = H_0^{(0)} = 0$.

A standard PR-averaged decomposition (cf. Section 4.1 below for the full derivation) yields

$$\sqrt{n}(\bar{\theta}_n^{(\alpha)} - \theta^*) = W + D_1,$$

where the leading martingale-like term is

$$W = - \frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} Q_l \epsilon(Z_l), \quad Q_l = \alpha \sum_{k=l}^{n-1} (I - \alpha \bar{A})^{n-k-1},$$

and the residual term is

$$D_1 = \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \Gamma_{1:k} (\theta_0 - \theta^*) + \frac{1}{\sqrt{n}} \sum_{k=1}^{n-1} J_k^{(0)}.$$

2.2. RR Combination and the $\tilde{J}_n^{(0,\alpha)}$ Term

Applying the decomposition of the previous subsection separately to step sizes α and 2α and forming the Richardson–Romberg combination, we obtain

$$\begin{aligned} \theta_n^{(\text{RR},\alpha)} - \theta^* &= \left[2\Gamma_{1:n}^{(\alpha)} - \Gamma_{1:n}^{(2\alpha)} \right] (\theta_0 - \theta^*) \\ &\quad + \left[2J_n^{(0,\alpha)} - J_n^{(0,2\alpha)} \right] + \left[2J_n^{(1,\alpha)} - J_n^{(1,2\alpha)} \right] + \left[2H_n^{(1,\alpha)} - H_n^{(1,2\alpha)} \right]. \end{aligned}$$

The first bracket is the deterministic transient and is exponentially small. The remaining three brackets are stochastic and need separate analysis. In this subsection we focus on the zeroth-order RR difference, which we denote

$$\tilde{J}_n^{(0,\alpha)} = 2J_n^{(0,\alpha)} - J_n^{(0,2\alpha)},$$

where, by definition,

$$J_n^{(0,\alpha)} = -\alpha \sum_{j=1}^n (I - \alpha \bar{A})^{n-j} \epsilon(Z_j).$$

Substituting and using the elementary identity $X^m - Y^m = (X - Y) \sum_{i=1}^m X^{i-1} Y^{m-i}$ with $X = I - \alpha \bar{A}$, $Y = I - 2\alpha \bar{A}$, $X - Y = \alpha \bar{A}$, we get

$$\begin{aligned}
\tilde{J}_n^{(0,\alpha)} &= -2\alpha \sum_{j=1}^n \left[(I - \alpha \bar{A})^{n-j} - (I - 2\alpha \bar{A})^{n-j} \right] \epsilon(Z_j) \\
&= -2\alpha^2 \bar{A} \sum_{j=1}^n \underbrace{\sum_{i=1}^{n-j} (I - \alpha \bar{A})^{i-1} (I - 2\alpha \bar{A})^{n-j-i}}_{=: H_j^{(n)}} \epsilon(Z_j) \\
&= -2\alpha^2 \bar{A} \sum_{j=1}^n H_j^{(n)} \epsilon(Z_j).
\end{aligned}$$

The extra factor $\alpha \bar{A}$ pulled out front is the source of the additional α -decay of the RR difference compared to a single LSA trajectory.

2.3. Norm Estimate for $H_j^{(n)}$

To bound $\|H_j^{(n)}\|$ we use the following standard result.

Lemma 1. Let $-\bar{A}$ be a Hurwitz matrix. Then for any $P = P^\top \succ 0$ there exists a unique $Q = Q^\top \succ 0$ satisfying

$$\bar{A}^\top Q + Q \bar{A} = P.$$

Moreover, letting

$$a := \frac{\lambda_{\min}(P)}{2 \|Q\|}, \quad \alpha_\infty := \min \left(\frac{\lambda_{\min}(P)}{2\kappa_Q \|\bar{A}\|_Q^2}, \frac{\|Q\|}{\lambda_{\min}(P)} \right),$$

with $\kappa_Q := \lambda_{\max}(Q)/\lambda_{\min}(Q)$, one has for all $\alpha \in [0, \alpha_\infty]$:

$$\alpha a \leq 1/2, \quad \|I - \alpha \bar{A}\|_Q^2 \leq 1 - \alpha a.$$

We now estimate $\|H_j^{(n)}\|$ by combining the triangle inequality, submultiplicativity, the equivalence $\|X\| \leq \kappa_Q^{1/2} \|X\|_Q$ (applied to **each** operator-norm factor, hence two factors of $\kappa_Q^{1/2}$ multiplying to κ_Q), and the Lyapunov contraction at step sizes α and 2α :

$$\begin{aligned}
\|H_j^{(n)}\| &\leq \sum_{i=1}^{n-j} \|I - \alpha \bar{A}\|^{i-1} \|I - 2\alpha \bar{A}\|^{n-j-i} && \text{(triangle + submult.)} \\
&\leq \kappa_Q \sum_{i=1}^{n-j} (1 - \alpha a)^{(i-1)/2} (1 - 2\alpha a)^{(n-j-i)/2} && \text{(equiv. + Lyapunov)} \\
&= \kappa_Q (1 - \alpha a)^{(n-j-1)/2} \sum_{k=0}^{n-j-1} \left(\frac{1 - 2\alpha a}{1 - \alpha a} \right)^{k/2} && \text{(reindex } k = n - j - i) \\
&\leq \kappa_Q (1 - \alpha a)^{(n-j-1)/2} \frac{1}{1 - \sqrt{\frac{1-2\alpha a}{1-\alpha a}}} && \text{(geometric series).}
\end{aligned}$$

For $\alpha a \leq 1/2$ the geometric rate is bounded below by an **elementary** inequality: combining $\sqrt{(1-2\alpha a)/(1-\alpha a)} \leq \sqrt{1-\alpha a}$ (since $(1-2\alpha a) \leq (1-\alpha a)^2$) with $1 - \sqrt{1-x} \geq x/2$ on $[0, 1]$ gives

$$1 - \sqrt{\frac{1-2\alpha a}{1-\alpha a}} \geq 1 - \sqrt{1-\alpha a} \geq \frac{\alpha a}{2}, \quad \text{i.e.} \quad \frac{1}{1 - \sqrt{\frac{1-2\alpha a}{1-\alpha a}}} \leq \frac{2}{\alpha a}.$$

Defining the chapter-local constant

$$\bar{C}_A := \kappa_Q$$

(distinct from the assumption-2 sup-norm constant C_A , which enters separately via $\|\bar{A}\|$ in the next subsection), we arrive at the final estimate

$$\|H_j^{(n)}\| \leq \bar{C}_A (1 - \alpha a)^{(n-j-1)/2} \frac{2}{\alpha a}.$$

The kernel decays geometrically in $n - j$ at rate $\sqrt{1 - \alpha a}$ but its summed weight is of order $1/(\alpha a)$. The next subsection shows how the prefactor α^2 in $\tilde{J}_n^{(0, \alpha)}$ exactly compensates this divergence.

2.4. L^2 Bound for the Zeroth-Order Term

The expression

$$\tilde{J}_n^{(0, \alpha)} = - \sum_{j=1}^n 2\alpha^2 \bar{A} H_j^{(n)} \epsilon(Z_j)$$

is a weighted sum of values of the centered noise function ϵ along the Markov chain. We control its tail via the following Markov concentration statement.

Lemma 2. Assume **UGE 1**. Let $\{g_i\}_{i=1}^n$ be a family of measurable functions $g_i : Z \rightarrow \mathbb{R}^d$ such that

$$c_i = \|g_i\|_\infty < \infty \quad \text{for all } i \geq 1, \quad \pi(g_i) = 0 \quad \text{for all } i \in \{1, \dots, n\}.$$

Then, for any initial distribution ξ on $(Z, \mathcal{Z})$, any $n \in \mathbb{N}$ and any $t \geq 0$,

$$\mathbb{P}_\xi \left(\left\| \sum_{i=1}^n g_i(Z_i) \right\| \geq t \right) \leq 2 \exp \left(- \frac{t^2}{2u_n^2} \right),$$

where

$$u_n = 8 \left(\sum_{i=1}^n c_i^2 \right)^{1/2} \sqrt{t_{\text{mix}}}.$$

Proof. The proof follows the lines of Durmus et al. (2025, Lemma 9).

We apply the lemma with $g_{j(z)} = -2\alpha^2 \bar{A} H_j^{(n)} \epsilon(z)$. Each g_j is centered under π since $\pi(\epsilon) = 0$, so the centering hypothesis holds. Combining the bound $\|\bar{A}\| \leq C_A$ (Assumption 2) with the previous subsection, define

$$\tilde{C}_A := C_A \bar{C}_A = C_A \kappa_Q.$$

Then the per-summand bound is

$$\begin{aligned} \|g_j\|_\infty &\leq 2\alpha^2 \tilde{C}_A (1 - \alpha a)^{(n-j-1)/2} \frac{2}{\alpha a} \|\epsilon\|_\infty \\ &= 4\alpha \tilde{C}_A \|\epsilon\|_\infty (1 - \alpha a)^{(n-j-1)/2}. \end{aligned}$$

Note the prefactor α^2 collapsing to α : the $1/(\alpha a)$ blow-up in $H_j^{(n)}$ is exactly absorbed by one factor of α . Squaring and summing over j , using $1 - \alpha a \geq 1/2$:

$$\begin{aligned} \sum_{j=1}^n \|g_j\|_\infty^2 &\leq 16\alpha^2 \tilde{C}_A^2 \|\epsilon\|_\infty^2 \sum_{j=1}^n (1 - \alpha a)^{n-j-1} \\ &\leq 16\alpha^2 \tilde{C}_A^2 \|\epsilon\|_\infty^2 \frac{1}{(1 - \alpha a)\alpha a} \\ &\leq \frac{32\alpha \tilde{C}_A^2 \|\epsilon\|_\infty^2}{a}. \end{aligned}$$

Plugging this into the variance proxy $u_n^2 = 64 t_{\text{mix}} \sum_{j=1}^n \|g_j\|_\infty^2$ from the lemma and defining

$$\hat{C}_A := \sqrt{2048} \tilde{C}_A \|\epsilon\|_\infty \sqrt{\frac{t_{\text{mix}}}{a}} = 32\sqrt{2} \tilde{C}_A \|\epsilon\|_\infty \sqrt{\frac{t_{\text{mix}}}{a}},$$

gives

$$u_n^2 \leq \hat{C}_A^2 \alpha.$$

To convert the sub-Gaussian tail into a moment bound, we use the following standard fact.

Lemma 3. Let X be an $\mathbb{R}^d$ -valued random variable satisfying

$$\mathbb{P}(\|X\| \geq t) \leq 2 \exp\left(-\frac{t^2}{2\sigma^2}\right) \quad \text{for all } t \geq 0,$$

for some $\sigma^2 > 0$. Then, for any $p \geq 2$, it holds that

$$\mathbb{E}[\|X\|^p] \leq 2 p^{p/2} \sigma^p.$$

Applying the lemma with $\sigma^2 = u_n^2 \leq \hat{C}_A^2 \alpha$ and an arbitrary $p \geq 2$ gives

$$\mathbb{E}^{1/p}[\|\tilde{J}_n^{(0,\alpha)}\|^p] \leq 2^{1/p} \sqrt{p} \hat{C}_A \sqrt{\alpha} \leq 2\sqrt{p} \hat{C}_A \sqrt{\alpha},$$

or equivalently

$$\frac{1}{\sqrt{\alpha}} \mathbb{E}^{1/p}[\|\tilde{J}_n^{(0,\alpha)}\|^p] \leq 2\sqrt{p} \hat{C}_A.$$

The zeroth-order RR difference is therefore $O(\sqrt{\alpha})$ in L^p for every $p \geq 2$, uniformly in n — an order- $\sqrt{\alpha}$ gain over a single LSA trajectory, which is $O(1)$ in L^p in the stationary regime. The $p = 2$ statement used in the rest of the analysis is a special case.

3. Last Iterate Analysis

3.1. Centered Bound for the Shifted First-Order Perturbation

The purpose of this section is to isolate the last-iteration weighted term which appears in the analysis of $J_n^{(1,\alpha)}$, and to record a clean L_p bound for its centered part. The proof uses the deterministic-product perturbation expansion from Samsonov et al. (2025, Proposition 9), specialized to the constant-stepsize setting.

Let

$$B = I - \alpha \bar{A}.$$

Throughout this section we assume $0 < \alpha \leq \alpha_\infty$, so that

$$\|B^m\|_Q \leq (1 - \alpha a)^{m/2}, \quad m \geq 0.$$

All constants denoted by C may depend on C_A , κ_Q , and norm equivalence constants, but not on α , n , p , or t_{mix} .

The deterministic-product components are defined by

$$J_n^{(0,\alpha)} = B J_{n-1}^{(0,\alpha)} - \alpha \epsilon(Z_n), \quad J_0^{(0,\alpha)} = 0,$$

and

$$J_n^{(1,\alpha)} = B J_{n-1}^{(1,\alpha)} - \alpha \tilde{A}(Z_n) J_{n-1}^{(0,\alpha)}, \quad J_0^{(1,\alpha)} = 0.$$

Hence

$$J_n^{(0,\alpha)} = -\alpha \sum_{k=1}^n B^{n-k} \epsilon(Z_k),$$

and

$$\begin{aligned} J_n^{(1,\alpha)} &= -\alpha \sum_{j=1}^n B^{n-j} \tilde{A}(Z_j) J_{j-1}^{(0,\alpha)} \\ &= \alpha^2 \sum_{1 \leq k < j \leq n} B^{n-j} \tilde{A}(Z_j) B^{j-1-k} \epsilon(Z_k) \\ &= \alpha^2 \sum_{k=1}^{n-1} \sum_{r=1}^{n-k} B^{n-k-r} \tilde{A}(Z_{k+r}) B^{r-1} \epsilon(Z_k). \end{aligned}$$

The sign in the second line is positive because $J_{j-1}^{(0,\alpha)}$ already contains the minus sign.

The last display is for $J_n^{(1,\alpha)}$ itself. The last-iteration estimate below operates on a **shifted** version, obtained by inserting one additional left factor B into the recursion:

$$S_n = \sum_{t=0}^{n-1} B^{n-t} \tilde{A}(Z_{t+1}) J_t^{(0,\alpha)}.$$

Reindexing $j = t + 1$ rewrites S_n in the same form as $J_n^{(1,\alpha)}$ but with one extra power of B :

$$S_n = \sum_{j=1}^n B^{n-j+1} \tilde{A}(Z_j) J_{j-1}^{(0,\alpha)} = -\frac{1}{\alpha} B J_n^{(1,\alpha)}.$$

Thus the corresponding shifted first-order contribution is

$$T_n^{(1,\alpha)} = -\alpha S_n = B J_n^{(1,\alpha)},$$

i.e. $T_n^{(1,\alpha)}$ is exactly $J_n^{(1,\alpha)}$ pre-multiplied by one additional B . In particular, any L_p bound on S_n (or $T_n^{(1,\alpha)}$) implies the same bound on $J_n^{(1,\alpha)}$ up to the contraction factor $\|B\|_Q \leq 1$.

Lemma 4. Assume the Markov chain is started from stationarity, that is, the law of Z_1 is π , and $\pi(\tilde{A}) = 0$. For every $p \geq 2$,

$$\|S_n - \mathbb{E}S_n\|_{L_p} \leq C \|\epsilon\|_\infty \left(p^{3/2} t_{\text{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\frac{\alpha}{a}} \right).$$

Consequently,

$$\|T_n^{(1,\alpha)} - \mathbb{E}T_n^{(1,\alpha)}\|_{L_p} \leq C\alpha \|\epsilon\|_\infty \left(p^{3/2} t_{\text{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\frac{\alpha}{a}} \right).$$

Proof. Since $J_0^{(0,\alpha)} = 0$, the term $t = 0$ in S_n vanishes. Substituting the explicit formula for $J_t^{(0,\alpha)}$ gives

$$\begin{aligned} S_n &= -\alpha \sum_{t=1}^{n-1} \sum_{k=1}^t B^{n-t} \tilde{A}(Z_{t+1}) B^{t-k} \epsilon(Z_k) \\ &= -\alpha \sum_{k=1}^{n-1} H_{k+1}^{(w)} \epsilon(Z_k), \end{aligned}$$

where, after the change of summation index $l = t - k + 1$,

$$H_{k+1}^{(w)} = \sum_{l=1}^{n-k} B^{n-k-l+1} \tilde{A}(Z_{k+l}) B^{l-1}.$$

The kernel $H_{k+1}^{(w)}$ acts on the past noise $\epsilon(Z_k)$ through future states Z_{k+l} , so $H_{k+1}^{(w)} \epsilon(Z_k)$ is a future-weighted bilinear functional of the trajectory.

Define

$$\mu_k^{(w)} = \mathbb{E}_\pi \left[H_{k+1}^{(w)} \epsilon(Z_k) \right].$$

Then

$$S_n - \mathbb{E}S_n = -\alpha \sum_{k=1}^{n-1} \left\{ H_{k+1}^{(w)} \epsilon(Z_k) - \mu_k^{(w)} \right\}.$$

Let $\mathcal{F}_k = \sigma(Z_1, \dots, Z_k)$. The Markov property gives

$$\mathbb{E} \left[H_{k+1}^{(w)} \epsilon(Z_k) \mid \mathcal{F}_k \right] = v_k^{(w,\epsilon)}(Z_k),$$

where

$$v_k^{(w)}(z) = \sum_{l=1}^{n-k} B^{n-k-l+1} (Q^l \tilde{A})(z) B^{l-1}, \quad v_k^{(w,\epsilon)}(z) = v_k^{(w)}(z) \epsilon(z).$$

Here Q denotes the one-step Markov transition kernel of $(Z_k)_{k \geq 1}$, acting on bounded matrix-valued functions by integration against the conditional law:

$$(Q\tilde{A})(z) = \int \tilde{A}(u) Q(z, du) = \mathbb{E}[\tilde{A}(Z_{k+1}) \mid Z_k = z],$$

and Q^l is its l -fold iterate, the l -step kernel $Q^l(z, du) = \mathbb{P}(Z_{k+l} \in du \mid Z_k = z)$, so that

$$(Q^l \tilde{A})(z) = \int \tilde{A}(u) Q^l(z, du) = \mathbb{E}[\tilde{A}(Z_{k+l}) \mid Z_k = z].$$

In particular $(Q^l \tilde{A})(z) \rightarrow \pi(\tilde{A}) = 0$ at the geometric rate dictated by UGE, which is the only fact about Q^l used below.

Under stationarity, $\mu_k^{(w)} = \pi(v_k^{(w, \varepsilon)})$. Therefore

$$S_n - \mathbb{E}S_n = -\alpha(U_M + U_R),$$

with

$$U_M = \sum_{k=1}^{n-1} \left\{ H_{k+1}^{(w)} \epsilon(Z_k) - v_k^{(w, \varepsilon)}(Z_k) \right\}$$

and

$$U_R = \sum_{k=1}^{n-1} \left\{ v_k^{(w, \varepsilon)}(Z_k) - \pi(v_k^{(w, \varepsilon)}) \right\}.$$

This decomposition splits the centered statistic into two structurally different parts: U_R is an ordinary centered additive functional of the original Markov chain, while U_M is a future-centered bilinear term — the conditional expectation given $\mathcal{F}_k$ vanishes summand-wise, but the summands are not forward martingale differences. We bound each piece separately.

Step 1: bound on U_R . Because $\pi(\tilde{A}) = 0$, applying Q^l followed by integration against π replaces $\tilde{A}$ by its l -step propagation away from stationarity. UGE then gives the geometric Dobrushin bound

$$\|(Q^l \tilde{A})(z)\| = \left\| \int \tilde{A}(u) \{Q^l(z, du) - \pi(du)\} \right\| \leq 2C_A \Delta(Q^l) \leq 2C_A (1/4)^{\lfloor l/t_{\text{mix}} \rfloor}.$$

Inserting this into $v_k^{(w)}$ and factoring out the slow rate $(1 - \alpha a)^{(n-k)/2}$ from the two B -powers, the inner sum becomes a fast-decaying l -series:

$$\begin{aligned} \|v_k^{(w)}\|_\infty &\leq C \sum_{l=1}^{n-k} (1 - \alpha a)^{(n-k-l+1)/2} \Delta(Q^l) (1 - \alpha a)^{(l-1)/2} \quad (\text{triangle + Lyapunov}) \\ &\leq C (1 - \alpha a)^{(n-k)/2} \sum_{l=1}^{\infty} (1/4)^{\lfloor l/t_{\text{mix}} \rfloor} \quad (\text{extend to } l = \infty) \\ &\leq C t_{\text{mix}} (1 - \alpha a)^{(n-k)/2} \quad (\text{geometric block sum}). \end{aligned}$$

Multiplying by $\|\epsilon\|_\infty$ gives

$$\|v_k^{(w, \varepsilon)}\|_\infty \leq C t_{\text{mix}} \|\epsilon\|_\infty (1 - \alpha a)^{(n-k)/2}.$$

The functions $v_k^{(w, \varepsilon)}(Z_k) - \pi(v_k^{(w, \varepsilon)})$ are centered under π and uniformly bounded by the previous display, so the weighted Markov concentration/Rosenthal bound for centered time-dependent functions yields

$$\begin{aligned}
\|U_R\|_{L_p} &\leq Cp^{1/2}t_{\text{mix}}^{1/2}\left(\sum_{k=1}^{n-1}\|v_k^{(w,\varepsilon)}\|_\infty^2\right)^{1/2} && \text{(Rosenthal)} \\
&\leq Cp^{1/2}t_{\text{mix}}^{3/2}\|\epsilon\|_\infty\left(\sum_{k=1}^{n-1}(1-\alpha a)^{n-k}\right)^{1/2} && \text{(plug previous bound)} \\
&\leq Cp^{1/2}t_{\text{mix}}^{3/2}\|\epsilon\|_\infty\frac{1}{\sqrt{\alpha a}} && \text{(geometric series).}
\end{aligned}$$

Step 2: bound on U_M . Project onto a deterministic unit vector u and unfold $H_{k+1}^{(w)}u$ as a sum over future states:

$$H_{k+1}^{(w)}u = \sum_{l=1}^{n-k} g_{k,l}(Z_{k+l}), \quad g_{k,l}(z) = B^{n-k-l+1}\tilde{A}(z)B^{l-1}u.$$

Each $g_{k,l}$ is centered under π (since $\pi(\tilde{A}) = 0$), and the two B -powers give the uniform bound

$$\|g_{k,l}\|_\infty \leq C(1-\alpha a)^{(n-k)/2},$$

which is independent of l . Squaring and summing over l produces only linear growth in $n-k$:

$$\sum_{l=1}^{n-k}\|g_{k,l}\|_\infty^2 \leq C(n-k)(1-\alpha a)^{n-k}.$$

Note the contrast with the U_R analysis: there, UGE folded the l -sum into a single t_{mix} factor, whereas here the same l -independent bound is applied $(n-k)$ times — the price of conditional centering being weaker than π -centering.

Applying the Markov concentration lemma to the future chain conditionally on $\mathcal{F}_k$,

$$\|H_{k+1}^{(w)}u - \mathbb{E}[H_{k+1}^{(w)}u | \mathcal{F}_k]\|_{L_p} \leq Cp^{1/2}t_{\text{mix}}^{1/2}\sqrt{(n-k)(1-\alpha a)^{n-k}},$$

and multiplying by $\epsilon(Z_k)$ (which is $\mathcal{F}_k$ -measurable and bounded by $\|\epsilon\|_\infty$) gives

$$\|H_{k+1}^{(w)}\epsilon(Z_k) - v_k^{(w,\varepsilon)}(Z_k)\|_{L_p} \leq Cp^{1/2}t_{\text{mix}}^{1/2}\|\epsilon\|_\infty\sqrt{(n-k)(1-\alpha a)^{n-k}}.$$

The summands in U_M are centered conditionally on $\mathcal{F}_k$, but they are *not* forward martingale differences: the kernel $H_{k+1}^{(w)}$ peeks into the future of Z_k . To assemble the per- k bounds into a bound on the sum we therefore invoke the weighted Burkholder/block-coupling estimate for future-centered bilinear Markov sums of Samsonov et al. (2025, Proposition 9), specialized to the present constant-stepsizes kernel:

$$\|U_M\|_{L_p} \leq Cp\left(\sum_{k=1}^{n-1}\|H_{k+1}^{(w)}\epsilon(Z_k) - v_k^{(w,\varepsilon)}(Z_k)\|_{L_p}^2\right)^{1/2}.$$

Plugging in the previous bound and evaluating the resulting sum:

$$\begin{aligned}
\|U_M\|_{L_p} &\leq Cp^{3/2}t_{\text{mix}}^{1/2}\|\epsilon\|_\infty\left(\sum_{k=1}^{n-1}(n-k)(1-\alpha a)^{n-k}\right)^{1/2} && \text{(plug previous bound)} \\
&\leq Cp^{3/2}t_{\text{mix}}^{1/2}\|\epsilon\|_\infty\frac{1}{\alpha a} && \left(\text{use } \sum_m m x^m \leq Cx^{-2} \right).
\end{aligned}$$

The extra factor $1/\sqrt{\alpha a}$ relative to U_R is precisely the cost of using conditional rather than stationary centering.

Step 3: assembly. Combining the two pieces via $S_n - \mathbb{E}S_n = -\alpha(U_M + U_R)$ and the triangle inequality,

$$\begin{aligned} \|S_n - \mathbb{E}S_n\|_{L_p} &\leq \alpha (\|U_M\|_{L_p} + \|U_R\|_{L_p}) \\ &\leq C \|\epsilon\|_\infty \left(p^{3/2} t_{\text{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\frac{\alpha}{a}} \right). \end{aligned}$$

The first term (from U_M) is the leading contribution for small α ; the second (from U_R) carries the heavier t_{mix} dependence but vanishes as $\alpha \rightarrow 0$. Multiplying by α gives the asserted bound for $T_n^{(1,\alpha)} = -\alpha S_n$. $\square$

Remark 1. The stationarity assumption is used only to avoid boundary notation in this local calculation. For a non-stationary initial law, UGE gives the same bulk estimate plus the standard exponentially small mixing boundary terms. The bias of $T_n^{(1,\alpha)}$ is a separate issue and is controlled through the stationary expansion of $\mathbb{E}J_n^{(1,\alpha)}$ in Levin et al. (2025).

3.2. Application to the PR-averaged RR Misadjustment

The PR-averaged Richardson–Romberg expansion produces, after Step (S8) of the Samsonov scheme applied separately at step sizes α and 2α , a “misadjustment” remainder

$$D_1^{\text{mis, RR}} = \frac{1}{\sqrt{n}} \sum_{k=n_0}^{n-1} \left(2J_k^{(1,\alpha)} - J_k^{(1,2\alpha)} \right),$$

whose centered part must be controlled to feed into a Berry–Esseen statement for $\sqrt{n}(\bar{\theta}_n^{(\text{RR},\alpha)} - \theta^*)$. The bias is not the obstruction: by Levin et al. (2025, Proposition 2),

$$\mathbb{E}_\pi [J_\infty^{(1,\alpha)}] = \alpha\Delta + O(\alpha^2),$$

so the linear term $\alpha\Delta$ cancels in the RR-combination and the stationary bias of $D_1^{\text{mis, RR}}$ is $O(\alpha^2)$. What remains is the centered fluctuation.

Define

$$\Phi(p, \alpha) := p^{3/2} \frac{t_{\text{mix}}^{1/2}}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\alpha/a}.$$

The lemma applied at α and at 2α gives, separately,

$$\|T_n^{(1,\alpha)} - \mathbb{E}T_n^{(1,\alpha)}\|_{L_p} \leq C\alpha \Phi(p, \alpha), \quad \|T_n^{(1,2\alpha)} - \mathbb{E}T_n^{(1,2\alpha)}\|_{L_p} \leq C\alpha \Phi(p, 2\alpha) \leq C'\alpha \Phi(p, \alpha),$$

where $C' = \sqrt{2}C$ absorbs the $\sqrt{2}$ -factor coming from the 2α scaling. Combining the two by the triangle inequality and using the index-shift identity $T_k^{(1,\alpha)} = B J_k^{(1,\alpha)}$ from the start of this section, we get

$$\left\| \left(2J_k^{(1,\alpha)} - J_k^{(1,2\alpha)} \right) - \mathbb{E} \left(2J_k^{(1,\alpha)} - J_k^{(1,2\alpha)} \right) \right\|_{L_p} \leq C\alpha \Phi(p, \alpha),$$

uniformly in k . PR-averaging through $\frac{\sqrt{n}}{n-n_0}$ and absorbing the constant therefore yields

$$\|D_1^{\text{mis, RR}}\|_{L_p} = O(\sqrt{n}\alpha).$$

At the optimal scale $\alpha \asymp n^{-1/2}$ this is $O(1)$, and as a remainder in a Berry–Esseen statement of order $n^{-1/4}$ it is **useless**: the misadjustment must be at least $o(1)$ to be subleading.

3.3. Conclusion: The Triangle Inequality Loses the RR Cancellation

Two structural reasons explain why the single- α centered bound proved above does not, by itself, deliver a usable RR-misadjustment estimate.

The triangle inequality is blind to coupling. Each $J_k^{(1,\alpha)}$ is genuinely $O(\alpha)$ in L_p , and the same is true at step 2α . Their RR-difference is therefore at most $O(\alpha)$ when bounded termwise. The cancellation that Richardson–Romberg actually provides — one extra factor of α on the difference of two processes coupled through the same noise realization $\{Z_k\}$ — is invisible to a $|2X - Y| \leq 2|X| + |Y|$ split.

The cancellation lives in the kernel. Reading the RR-difference as a single weighted sum,

$$T_n^{(1,\alpha)} - T_n^{(1,2\alpha)} = - \sum_{k=1}^{n-1} \mathcal{H}_{k+1}^{\text{RR}} \epsilon(Z_k),$$

with

$$\mathcal{H}_{k+1}^{\text{RR}} = \sum_{l=1}^{n-k} \left(2\alpha B_\alpha^{n-k-l+1} \tilde{A}(Z_{k+l}) B_\alpha^{l-1} - 2\alpha B_{2\alpha}^{n-k-l+1} \tilde{A}(Z_{k+l}) B_{2\alpha}^{l-1} \right),$$

the elementary identity $X^m - Y^m = (X - Y) \sum_{i=1}^m X^{i-1} Y^{m-i}$ with $X = B_\alpha$, $Y = B_{2\alpha}$, $X - Y = \alpha \bar{A}$ extracts an additional factor of α uniformly in the kernel — the very same mechanism that gave the $\sqrt{\alpha}$ gain for the zeroth-order RR-difference $\tilde{J}_n^{(0,\alpha)}$ in the previous chapter. Repeating the conditional/ stationary centering argument of the present section with $\mathcal{H}^{\text{RR}}$ in place of $H^{(w)}$ would consequently gain one factor of α globally, lifting the misadjustment estimate to

$$\|D_1^{\text{mis, RR}}\|_{L_p} = O(\sqrt{n} \alpha^2),$$

i.e. $O(n^{-1/2})$ at $\alpha \asymp n^{-1/2}$ — strictly below the martingale Berry–Esseen scale $n^{-1/4}$.

Why we stop here. For a first Berry–Esseen statement of rate $n^{-1/4}$, the RR-kernel refinement is not actually needed: the same final rate already follows from the centered bound on the unweighted statistic $\sum \tilde{A}(Z_{t+1}) J_t^{(0,\alpha)}$ in Levin et al. (2025, Proposition 3 / Corollary 6), combined with the depth-two perturbation expansion $H^{(1)} = J^{(2)} + H^{(2)}$ and the high-order moment bounds on $J^{(2)}$, $H^{(2)}$ (Levin et al., 2025, Propositions 8–9). Inserted into the RR-decomposition of equation (26) of that paper, these tools give a remainder of order $n^{-1/4}$, which matches the leading martingale Berry–Esseen contribution.

The lemma proved in this section is therefore a clean **intermediate object**: it isolates the shifted last-iteration statistic S_n , decomposes its centered part into an additive functional U_R and a future-centered bilinear part U_M , and exposes the conditional-centering split that any finer RR-analysis must respect. It does not on its own improve the leading order of the Berry–Esseen bound. A genuine improvement — pushing the misadjustment strictly below $n^{-1/4}$ — would require lifting the shifted-kernel argument from a single α to the RR-difference $\nabla_\alpha S_n$, with the kernel $\mathcal{H}^{\text{RR}}$ above. Because the additional α -factor is purely deterministic (it lives entirely in $B_\alpha^m - B_{2\alpha}^m$), the conditional-centering, weighted Burkholder and Markov–Rosenthal pieces of the proof carry over verbatim. We record this RR-kernel version as the principal open thread of the present analysis rather than expand on it here.

4. Richardson–Romberg PR Weight Bounds

4.1. PR-Averaged Error Decomposition and the RR Weight

The starting point for all subsequent estimates is the **explicit** representation of $\sqrt{n}(\bar{\theta}_n^{(\text{RR}, \alpha)} - \theta^*)$ as a noise-weighted sum with a deterministic kernel. This subsection derives that representation step by step.

Depth-one expansion. Unfolding the error recursion of Chapter 1 gives, for every $k \geq 1$,

$$\theta_k^{(\alpha)} - \theta^* = -\alpha \sum_{l=1}^k \Gamma_{l+1:k}^{(\alpha)} \epsilon(Z_l) + \Gamma_{1:k}^{(\alpha)} (\theta_0 - \theta^*), \quad \Gamma_{l+1:k}^{(\alpha)} := \prod_{j=l+1}^k (I - \alpha A(Z_j)).$$

Replacing each random product $\Gamma_{l+1:k}^{(\alpha)}$ by its deterministic counterpart $B_\alpha^{k-l} := (I - \alpha \bar{A})^{k-l}$ and absorbing the difference into a higher-order remainder yields the **depth-one decomposition** (Samsonov et al., 2025, Proposition 9):

$$\theta_k^{(\alpha)} - \theta^* = -\alpha \sum_{l=1}^k B_\alpha^{k-l} \epsilon(Z_l) + B_\alpha^k (\theta_0 - \theta^*) + R_k^{(\alpha)},$$

where $R_k^{(\alpha)} := J_k^{(1, \alpha)} + H_k^{(1, \alpha)}$ collects all contributions of order $\alpha^{3/2}$ or higher in L_p . The first sum is the **depth-zero** term and is the only piece that carries the limiting Gaussian.

PR averaging produces $Q_l^{(\alpha)}$. Recall the PR average $\bar{\theta}_n^{(\alpha)} = (n - n_0)^{-1} \sum_{k=n_0}^{n-1} \theta_k^{(\alpha)}$. For notational clarity we set $n_0 = 0$ from this point on (the burn-in adds only an exponentially small transient). Subtracting θ^* , substituting the depth-one decomposition above, and **exchanging the order of summation** in the depth-zero piece,

$$\sum_{k=0}^{n-1} \sum_{l=1}^k B_\alpha^{k-l} \epsilon(Z_l) = \sum_{l=1}^{n-1} \left(\sum_{k=l}^{n-1} B_\alpha^{k-l} \right) \epsilon(Z_l),$$

isolates each noise sample $\epsilon(Z_l)$ together with the deterministic **PR weight**

$$Q_l^{(\alpha)} := \alpha \sum_{k=l}^{n-1} B_\alpha^{k-l} = \alpha \sum_{j=0}^{n-l-1} B_\alpha^j.$$

Operationally, $Q_l^{(\alpha)}$ collects the cumulative contribution of $\epsilon(Z_l)$ to all averaged iterates $\theta_l, \theta_{l+1}, \dots, \theta_{n-1}$. Multiplying the resulting identity by $\sqrt{n}$ gives the PR-averaged decomposition

$$\sqrt{n}(\bar{\theta}_n^{(\alpha)} - \theta^*) = W^{(\alpha)} + D^{(\alpha)},$$

where the **leading martingale-like sum** is

$$W^{(\alpha)} := -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} Q_l^{(\alpha)} \epsilon(Z_l),$$

and the remainder $D^{(\alpha)}$ packs the deterministic transient $D_{\text{tr}}^{(\alpha)}$ and the higher-order stochastic part $D_R^{(\alpha)}$:

$$D^{(\alpha)} := D_{\text{tr}}^{(\alpha)} + D_R^{(\alpha)}, \quad D_{\text{tr}}^{(\alpha)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} B_\alpha^k (\theta_0 - \theta^*), \quad D_R^{(\alpha)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} R_k^{(\alpha)}.$$

The first sum is a deterministic geometric tail and becomes exponentially small after a logarithmic burn-in $n_0 \asymp \log(n)/(a\alpha)$ (Lyapunov contraction $\|B_\alpha^{n_0}\|_Q \leq e^{-1}$). The second is the source of the leading non-Gaussian correction in the Berry–Esseen rate; its RR-combination $2D_R^{(\alpha)} - D_R^{(2\alpha)}$ is exactly the **misadjustment** $D_1^{\text{mis, RR}}$ controlled in Chapters 2 and 3.

RR combination produces $\mathcal{Q}_l^{\text{RR}}$. The Richardson–Romberg iterate $\bar{\theta}_n^{(\text{RR}, \alpha)} := 2\bar{\theta}_n^{(\alpha)} - \bar{\theta}_n^{(2\alpha)}$ inherits the PR decomposition **by linearity**: applying the previous display at step sizes α and 2α separately and combining,

$$\sqrt{n} \left(\bar{\theta}_n^{(\text{RR}, \alpha)} - \theta^* \right) = W^{\text{RR}} + D^{\text{RR}},$$

$$W^{\text{RR}} := 2W^{(\alpha)} - W^{(2\alpha)} = -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} \mathcal{Q}_l^{\text{RR}} \epsilon(Z_l),$$

where the **RR weight** is

$$\mathcal{Q}_l^{\text{RR}} := 2Q_l^{(\alpha)} - Q_l^{(2\alpha)}.$$

Crucially, both PR averages share the **same** noise realization $\{Z_k\}$, so the bracket $\mathcal{Q}_l^{\text{RR}}$ is a single deterministic matrix kernel: all the stochastic content of W^{RR} now lives in $\epsilon(Z_l)$ alone.

Why $\mathcal{Q}_l^{\text{RR}}$ is the right object. Two questions about W^{RR} drive the rest of the chapter, and both answer themselves in terms of $\mathcal{Q}_l^{\text{RR}}$:

1. **Variance comparison.** The CLT identifies the limiting covariance of W^{RR} as $\Sigma_\infty = \bar{A}^{-1} \Sigma_\epsilon^{(\text{M})} \bar{A}^{-\top}$ (the Markov-chain CLT covariance), which is exactly what one obtains if every $\mathcal{Q}_l^{\text{RR}}$ is replaced by the asymptotic weight $\bar{A}^{-1}$. The finite- n deviation is therefore controlled by $\sum_l \|\mathcal{Q}_l^{\text{RR}} - \bar{A}^{-1}\|^2$ — see Section 4.5.
2. **Poisson-equation / Abel-summation remainder.** The standard Berry–Esseen route for Markov-chain noise solves the Poisson equation $\hat{\epsilon} - Q\hat{\epsilon} = \epsilon$, replaces $\epsilon(Z_l)$ by $\hat{\epsilon}(Z_l) - \hat{\epsilon}(Z_{l+1})$ (up to a martingale increment), and Abel-sums against the weight sequence $\mathcal{Q}_l^{\text{RR}}$. The resulting remainder has norm bounded by the **total variation** $\sum_l \|\mathcal{Q}_{l+1}^{\text{RR}} - \mathcal{Q}_l^{\text{RR}}\|$.

Both quantities are estimated in Sections 4.3–4.4 below. The closed-form identities of Section 4.2 reduce them to elementary operator-norm calculations on $B_\alpha^m - B_{2\alpha}^m$.

4.2. Closed-Form Identities

Throughout this chapter write

$$B_\alpha := I - \alpha \bar{A}, \quad B_{2\alpha} := I - 2\alpha \bar{A}, \quad k := n - l.$$

We assume $\alpha, 2\alpha \in (0, \alpha_\infty]$, so that the Lyapunov contraction $\|B_\alpha^m\|_Q^2 \leq (1 - \alpha a)^m$ and $\|B_{2\alpha}^m\|_Q^2 \leq (1 - 2\alpha a)^m$ hold (and we use freely $1 - 2\alpha a \leq 1 - \alpha a$).

The geometric-series identity $\sum_{j=0}^{m-1} B_\alpha^j = (\alpha \bar{A})^{-1} (I - B_\alpha^m)$ converts into the closed form

$$Q_l^{(\alpha)} = \alpha \sum_{j=0}^{n-l-1} B_\alpha^j = \alpha (\alpha \bar{A})^{-1} (I - B_\alpha^{n-l}) = \bar{A}^{-1} (I - B_\alpha^{n-l}).$$

This makes the convergence $Q_l^{(\alpha)} \rightarrow \bar{A}^{-1}$ as $k = n - l \rightarrow \infty$ explicit: the only obstruction is the geometric Lyapunov tail B_α^k . Two immediate consequences are

$$Q_l^{(\alpha)} - \bar{A}^{-1} = -\bar{A}^{-1} B_\alpha^{n-l}, \quad Q_{l+1}^{(\alpha)} - Q_l^{(\alpha)} = -\alpha B_\alpha^{n-l-1}.$$

The first is the **asymptotic-weight error** and decays geometrically in k . The second is the **discrete derivative** used in Abel summation; note the explicit factor α in front.

Applying both identities at α and 2α and combining yields the basic **RR identities**:

$$\mathcal{Q}_l^{\text{RR}} - \bar{A}^{-1} = -\bar{A}^{-1} (2B_\alpha^k - B_{2\alpha}^k),$$

$$\mathcal{Q}_{l+1}^{\text{RR}} - \mathcal{Q}_l^{\text{RR}} = -2\alpha (B_\alpha^{k-1} - B_{2\alpha}^{k-1}).$$

These are the exact expressions bounded in the next subsection. Note the structural asymmetry between them:

- The asymptotic-weight error $2B_\alpha^k - B_{2\alpha}^k$ is at the **single-trajectory** rate $(1 - \alpha a)^{k/2}$ — the triangle inequality $|2X - Y| \leq 2|X| + |Y|$ does not see the RR coupling, and indeed the leading $-\bar{A}^{-1}$ in $\mathcal{Q}_l^{\text{RR}}$ is the same as in the single-step weight $Q_l^{(\alpha)}$.
- The discrete derivative $B_\alpha^{k-1} - B_{2\alpha}^{k-1}$, however, is a **true** difference of contractions evaluated at the same exponent. The elementary identity $X^m - Y^m = (X - Y) \sum_{i=1}^m X^{i-1} Y^{m-i}$ with $X = B_\alpha$, $Y = B_{2\alpha}$, $X - Y = \alpha \bar{A}$ extracts an extra factor of α , gaining one full power of α over the single-step case $Q_{l+1}^{(\alpha)} - Q_l^{(\alpha)} = -\alpha B_\alpha^{k-1}$.

This local α -gain in the discrete derivative is the **only** manifestation of Richardson–Romberg cancellation visible at the level of the PR weights.

4.3. Pointwise Bounds for the RR Weights

Lemma 5. Let $\alpha, 2\alpha \in (0, \alpha_\infty]$, set $C_Q := \kappa_Q^{1/2} \|\bar{A}^{-1}\|$ and $\tilde{C}_A := \kappa_Q \|\bar{A}\|$, and write $k = n - l$.

(i) For every $1 \leq l \leq n - 1$,

$$\|\mathcal{Q}_l^{\text{RR}} - \bar{A}^{-1}\| \leq 3C_Q(1 - \alpha a)^{k/2}.$$

(ii) For every $1 \leq l \leq n - 2$,

$$\|\mathcal{Q}_{l+1}^{\text{RR}} - \mathcal{Q}_l^{\text{RR}}\| \leq 2\tilde{C}_A \alpha^2 (k - 1) (1 - \alpha a)^{(k-2)/2}.$$

Proof of (i). Take norms in the basic RR identity above. Using the equivalence $\|\cdot\| \leq \kappa_Q^{1/2} \|\cdot\|_Q$ and the Lyapunov contraction at both step sizes,

$$\|2B_\alpha^k - B_{2\alpha}^k\|_Q \leq 2\|B_\alpha^k\|_Q + \|B_{2\alpha}^k\|_Q \leq 2(1 - \alpha a)^{k/2} + (1 - 2\alpha a)^{k/2} \leq 3(1 - \alpha a)^{k/2}.$$

Submultiplicativity in $\|\cdot\|_Q$ followed by norm equivalence then gives

$$\|\mathcal{Q}_l^{\text{RR}} - \bar{A}^{-1}\| \leq \kappa_Q^{1/2} \|\bar{A}^{-1}\| \cdot \|2B_\alpha^k - B_{2\alpha}^k\|_Q \leq 3C_Q(1 - \alpha a)^{k/2}.$$

Proof of (ii). Apply the elementary identity $X^m - Y^m = (X - Y) \sum_{i=1}^m X^{i-1} Y^{m-i}$ with $X = B_\alpha$, $Y = B_{2\alpha}$, $X - Y = \alpha \bar{A}$, and $m = k - 1$:

$$B_\alpha^{k-1} - B_{2\alpha}^{k-1} = \alpha \bar{A} \sum_{i=1}^{k-1} B_\alpha^{i-1} B_{2\alpha}^{k-1-i}.$$

Each summand obeys, by submultiplicativity in $\|\cdot\|_Q$ and Lyapunov contraction,

$$\begin{aligned} \|B_\alpha^{i-1} B_{2\alpha}^{k-1-i}\| &\leq \kappa_Q^{1/2} (1 - \alpha a)^{(i-1)/2} (1 - 2\alpha a)^{(k-1-i)/2} \\ &\leq \kappa_Q^{1/2} (1 - \alpha a)^{(k-2)/2}, \end{aligned}$$

where in the last step we used $1 - 2\alpha a \leq 1 - \alpha a$. Summing over $i \in \{1, \dots, k - 1\}$ and absorbing constants,

$$\|B_\alpha^{k-1} - B_{2\alpha}^{k-1}\| \leq \alpha \kappa_Q \|\bar{A}\| (k - 1) (1 - \alpha a)^{(k-2)/2}.$$

Inserting this bound into the discrete-difference identity finishes the proof. $\square$

4.4. Summed Bounds and Comparison with the Single-Step Case

Corollary 1. Under the assumptions of the previous lemma, uniformly in $n \geq 2$,

$$\sum_{l=1}^{n-1} \|Q_l^{\text{RR}} - \bar{A}^{-1}\|^2 \leq \frac{C_1}{\alpha a}, \quad \sum_{l=1}^{n-2} \|Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}\| \leq \frac{C_2}{a^2},$$

with $C_1 = 9C_Q^2$ and $C_2 = 32\tilde{C}_A$.

Proof. For the first sum apply (i) and the geometric series $\sum_{k \geq 1} (1 - \alpha a)^k \leq \frac{1}{\alpha a}$. For the second, apply (ii) and

$$\sum_{k \geq 2} (k-1) (1 - \alpha a)^{(k-2)/2} = \sum_{m \geq 0} (m+1) (1 - \alpha a)^{m/2} \leq \frac{1}{(1 - \sqrt{1 - \alpha a})^2} \leq \frac{4}{(\alpha a)^2},$$

where the last step uses $1 - \sqrt{1 - \alpha a} \geq \alpha \frac{a}{2}$ for $\alpha a \leq 1/2$. Multiplying by $2\tilde{C}_A \alpha^2$ gives the claim, up to the stated universal constant. $\square$

Remark 2. Comparison with the single-step case. The same identities for $Q_l^{(\alpha)}$ give $\|Q_l^{(\alpha)} - \bar{A}^{-1}\| \leq C_Q (1 - \alpha a)^{k/2}$ and $\|Q_{l+1}^{(\alpha)} - Q_l^{(\alpha)}\| \leq \alpha \kappa_Q^{1/2} (1 - \alpha a)^{(k-1)/2}$. Part (i) of the lemma above is therefore the same order as the single-step bound: the leading $-\bar{A}^{-1}$ term of Q_{n-1}^{RR} is exactly $-\bar{A}^{-1}$ (since $Q_{n-1}^{\text{RR}} = 0$ identically), so no factor of α is gained on the boundary. The Richardson–Romberg cancellation is visible in the successive difference of (ii): for fixed lag k the discrete derivative gains an extra factor of α . This local gain comes with the factor $k-1$, so after summing over the geometric tail it should not be read as a global extra factor of α in the Abel term. What is needed for the Poisson-equation remainder is the uniform total-variation bound in the corollary. On the variance side the corollary gives $\sum \|Q_l^{\text{RR}} - \bar{A}^{-1}\|^2 = O(\frac{1}{\alpha a})$, which yields the finite-horizon comparison error $O(\frac{1}{n\alpha a})$.

4.5. Variance Comparison

After the Poisson decomposition the leading martingale W^{RR} has predictable quadratic variation that converges to

$$\Sigma_n^{\text{RR}} := \frac{1}{n} \sum_{l=1}^{n-1} Q_l^{\text{RR}} \Sigma_\epsilon^{(\text{M})} (Q_l^{\text{RR}})^\top, \quad \sigma_n^{2, \text{RR}}(u) := u^\top \Sigma_n^{\text{RR}} u,$$

with $\Sigma_\epsilon^{(\text{M})}$ the long-run noise covariance of the Markov chain, $\Sigma_\epsilon^{(\text{M})} = \mathbb{E}_\pi [\epsilon(Z_0) \epsilon(Z_0)^\top] + 2 \sum_{j \geq 1} \mathbb{E}_\pi [\epsilon(Z_0) \epsilon(Z_j)^\top]$. The asymptotic covariance is

$$\Sigma_\infty = \bar{A}^{-1} \Sigma_\epsilon^{(\text{M})} \bar{A}^{-\top}, \quad \sigma^2(u) = u^\top \Sigma_\infty u.$$

The bounds of the previous section give a quantitative comparison between Σ_n^{RR} and Σ_∞ .

Lemma 6. Let $\alpha, 2\alpha \in (0, \alpha_\infty]$, set $\Sigma := \Sigma_\epsilon^{(\text{M})}$, and assume $\|\Sigma\| < \infty$. Then

$$\|\Sigma_n^{\text{RR}} - \Sigma_\infty\| \leq \frac{C_3}{n\alpha a},$$

with $C_3 = 12 C_Q \|\bar{A}^{-1}\| \|\Sigma\| + 9 C_Q^2 \|\Sigma\|$. Consequently, for every $u \in \mathbb{R}^d$,

$$|\sigma_n^{2, \text{RR}}(u) - \sigma^2(u)| \leq \frac{C_3 \|u\|^2}{n\alpha a}.$$

At the working scale $\alpha = c n^{-1/2}$ this is $O(n^{-1/2})$.

Proof. Write $\Delta_l := Q_l^{\text{RR}} - \bar{A}^{-1}$ and expand

$$Q_l^{\text{RR}} \Sigma (Q_l^{\text{RR}})^\top - \Sigma_\infty = \underbrace{\Delta_l \Sigma \bar{A}^{-\top} + \bar{A}^{-1} \Sigma \Delta_l^\top}_{R_{1,l}} + \underbrace{\Delta_l \Sigma \Delta_l^\top}_{R_{2,l}}.$$

Submultiplicativity of the operator norm yields the pointwise bounds

$$\| R_{1,l} \| \leq 2 \| \bar{A}^{-1} \| \| \Sigma \| \| \Delta_l \|, \quad \| R_{2,l} \| \leq \| \Sigma \| \| \Delta_l \|^2.$$

Summing the linear part with the bound from part (i) of the previous lemma and the geometric series $\sum_{k \geq 1} (1 - \alpha a)^{k/2} \leq \frac{1}{1 - \sqrt{1 - \alpha a}} \leq \frac{2}{\alpha a}$ (valid for $\alpha a \leq 1/2$),

$$\sum_{l=1}^{n-1} \| \Delta_l \| \leq 3C_Q \sum_{k=1}^{n-1} (1 - \alpha a)^{k/2} \leq \frac{6C_Q}{\alpha a}.$$

The quadratic part is bounded directly by the previous corollary,

$$\sum_{l=1}^{n-1} \| \Delta_l \|^2 \leq \frac{9C_Q^2}{\alpha a}.$$

Combining,

$$\begin{aligned} \| \Sigma_n^{\text{RR}} - \Sigma_\infty \| &\leq \frac{1}{n} \sum_{l=1}^{n-1} (\| R_{1,l} \| + \| R_{2,l} \|) \\ &\leq \frac{1}{n} \left(2 \| \bar{A}^{-1} \| \| \Sigma \| \cdot \frac{6C_Q}{\alpha a} + \| \Sigma \| \cdot \frac{9C_Q^2}{\alpha a} \right) = \frac{C_3}{n \alpha a}, \end{aligned}$$

which proves the operator-norm bound. The scalar bound on $|\sigma_n^{2, \text{RR}}(u) - \sigma^2(u)| = |u^\top (\Sigma_n^{\text{RR}} - \Sigma_\infty)u|$ follows from the Cauchy–Schwarz inequality. $\square$

Remark 3. The constant C_3 is independent of α and n ; only a enters through the Lyapunov contraction. The bound is tight in the geometric scale: the dominant contribution comes from the near-boundary region $l \approx n$, where $\| \Delta_l \|$ is bounded below by a positive constant (see the boundary identity $\mathcal{Q}_{n-1}^{\text{RR}} = 0$). The same calculation for the single-step PR weights yields $\| \Sigma_n^{\text{PR}} - \Sigma_\infty \| \leq \frac{C}{n \alpha a}$, so Richardson–Romberg does not improve the variance comparison rate, but it also does not degrade it: the leading-order variance is preserved.